

PRINCE RALAMBOMANARIVO

Software Engineer | Systems Architect | AI Researcher

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PROFESSIONAL SUMMARY

Software Engineer and Systems Architect with 6+ years of experience designing, building, and shipping production software across healthcare, AI, and systems infrastructure. Architected enterprise-scale data pipelines, agentic AI systems with planner/tool-loop execution, and cross-platform desktop applications compiled from Rust to WebAssembly. Maintainer of 40+ open-source projects. Strong track record of owning features end-to-end in HIPAA-regulated environments, from database optimization to deployment.

EXPERIENCE

Software Developer

May 2022 – Present

Medical Informatics Engineering

Fort Wayne, IN

- Architect and ship features end-to-end across healthcare platforms serving enterprise clients — patient records, immunization registries, occupational health, and compliance reporting modules
- Design and optimize complex SQL pipelines against MariaDB/MySQL databases, reducing query latency on reporting workflows handling millions of medical records
- Build and maintain full-stack applications (TypeScript, JavaScript, modern frameworks) powering clinical data entry, form rendering, and cross-system integrations
- Own developer tooling and workflow automation — built internal CLI tools to streamline Git/PR workflows and accelerate team velocity across Azure DevOps
- Enforce HIPAA compliance at the code level; conduct security-aware code reviews and maintain annual InfoSec certifications

Software Developer & IT Architect

Jan 2020 – Jan 2025

Rockdiva Nails Salon & Spa

Fort Wayne, IN

- Architected full web presence from scratch — booking system, responsive site, and backend infrastructure
- Managed network architecture, server provisioning, and ongoing system administration

Software Engineering Intern

May 2021 – Aug 2021

Parkview Mirro Center for Research and Innovation

Fort Wayne, IN

- Designed and built a secure web application integrating Twilio APIs to facilitate real-time communication between clinicians and patients
- Shipped production-quality code within regulated medical software development lifecycles

Software Developer & Multimedia Engineer

Jan 2019 – May 2022

Purdue University Fort Wayne – StudioM

Fort Wayne, IN

- Built internal tools and automated production workflows for university media departments
- Developed 90+ After Effects CEP extension tools for motion graphics pipelines

TECHNICAL SKILLS

Languages: Rust, TypeScript, Python, C, C++, Zig, Go, Java, C#, SQL, PHP, Dart, HTML/CSS

AI & ML: Agentic AI Systems, PyTorch, HuggingFace, LLM Integration (OpenAI, Anthropic, Ollama), LangChain, RAG, MCP, Whisper ASR, Fine-Tuning

Web & Apps: Svelte 5, React, Vue, Next.js, Angular, Tauri 2, Electron, Laravel, Django, Flask, Axum, Node.js, Vite, WebGL

Systems: Rust/WASM (wasm-pack), Wayland, Linux Kernel, Buildroot, Custom OS Images, x86_64 Bare-Metal, FPGA, GPU/CUDA

Data & Infra: MySQL, MariaDB, MongoDB, Firebase, SurrealDB, Docker, Kubernetes, Git, Azure DevOps, GitHub Actions, CI/CD

Embedded: ESP32, RP2040/RP2350, ESP-IDF, BLE, SPI/I2C, KiCad, Altium, PCB Design

EDUCATION

Bachelor of Science in Computer Science

May 2022

Purdue University, Fort Wayne

CERTIFICATIONS, AWARDS & LANGUAGES

- C & C++ Certification
- HIPAA Privacy & Security (annual)
- Security Awareness Training (ISAT)
- HealthCare InfoSec & Privacy Practitioner
- Outstanding Leadership Award – PFW
- Certificate of Merit – ELS

English (Bilingual) | **French** (Bilingual) | **Malagasy** (Native)